



Original research

Did injury incidence in alpine ski racing change after equipment regulations? An evaluation based on the injury surveillance system of the Austrian Ski Federation

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ARTICLE INFO

Article history:

Received 30 November 2019

Received in revised form 25 June 2020

Accepted 13 July 2020

Available online 24 July 2020

Keywords:

Skiing

Severe injuries

Knee

Anterior cruciate ligament

Equipment changes

ABSTRACT

Objectives: To assess the effects of alpine competition equipment regulations from 2003, 2007 and 2012 on severe injury incidence.

Design: Case study

Method: Data originated from records of the injury surveillance system of the Austrian Ski Federation. Injuries from the seasons 2001–2017 were divided in four periods between the equipment regulations. For comparison of consecutive periods, risk ratios (RR = later period / preceding period) with 95% CI were calculated. Total severe injury events, events with severe knee injuries, and events with severe ACL injuries were separately investigated.

Results: A significant increase of total severe injury incidence was found after the equipment regulation in 2003 (RR 1.52, 95% CI 1.00–2.31). None of the other comparisons revealed significance ($p < 0.05$) or statistical trends ($p < 0.1$). Only the minority (40%) of the RR showed a reduction in the injury incidence of the Austrian Ski Team (lowest RR 0.78). 60% of the RR increased after the regulations (highest RR 1.63).

Conclusions: Even though statistical uncertainties remain, our findings allow the conclusion, that the implemented equipment regulations did not cause a noticeable reduction of injuries. The three analysed equipment regulations were not appropriate or were counteracted by other factors.

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Practical Implications

- The probability of detecting true effects was restricted by the low sample size. Additionally, the Austrian Ski Team may have some limitations as representative sample for the FIS alpine ski racers. Thus, international collaborations for future data collection have to be established to describe injury risk in ski racing.
- Preventive measures should not only focus on equipment but address the whole spectrum of risk factors both internal and external to the athlete.
- Computer simulation models have to be developed to study risk factors systematically.

1. Introduction

Alpine ski racing is a spectacular and fascinating sport. Unfortunately, there are a concerning number of injuries. The overall injury incidence for Alpine World Cup ski racers exceed 30 injuries per 100 athletes per season,^{1,2} with more than one third relating to a time loss in training or competition of 28 days and more (severe injuries).^{3,4} Studies agree that the knee is the most frequently affected body part accounting for up to 40.7% of all reported injuries,^{2,5,6} with the anterior cruciate ligament (ACL) in the knee accounting for 13.6% of all injuries.⁴

In attempts to reduce injuries, the International Ski Federation (FIS) modified the equipment requirements every three to five years over the last two decades (2000, 2003, 2007, 2012 and 2017). These equipment regulations affected ski length, ski width at waist and shovel, ski side cut radius as well as standing height (distance between ski base and heel).⁷ Different dimensional characteristics were implemented for slalom, giant slalom, super-G and

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downhill; the key differences between the disciplines are speed and radii in the turns that increase sequentially from slalom (~45 km/h, 7 m) to downhill (~85 km/h, 40 m). Consequently, ski racers use the shortest and most shaped skis in slalom and the longest and least shaped in downhill. Previous work has shown that ski geometry can influence stability. Higher speed demands longer skis because the terrain absorbing function of the shovel is needed for skiing stability.⁸ Low side cut radii enhance the self-steering effect of skis and allow carving tight turn radii.⁹ However, this self-steering effect makes skis less controllable when skier speed is high. Standing height effects the ease of edging and makes the edge angle less controllable when too high.

To the best of our knowledge there is only one study investigating the effects of the FIS equipment regulations on injury incidence.³ In this study, six seasons before and three seasons after the equipment regulation of 2012 were analysed. No differences were found for severe injury incidence, knee injury incidence, or ACL injury incidence. However, injury incidence with a time loss of > 1 day and incidence of upper body injuries were lower in the seasons after the equipment regulation. This investigation has some limitations: (1) the study's sample consisted of Alpine World Cup ski racers only; (2) only injuries occurring during the Alpine World Cup competition season were recorded; and, (3) the analysis included season 2007 in which a different equipment regulation was valid when compared to the other five seasons prior to 2012.³

The purpose of this study was to assess the effect of the FIS equipment regulations of the years 2003, 2007 and 2012 on severe injury incidence. We hypothesized that injury incidence changed between the pre and post regulation periods.

2. Methods

The Austrian Ski Federation implemented an injury surveillance system in 1993. Until 2009 data was collected by a paper and pencil questionnaire; after 2009 an adapted online version was used. The following information was obtained for each incident: the medical report, questions for basic athlete data, and information about the injury such as affected body part, injury type and mechanism, environmental data, and subjective influencing factors. The specific diagnosis and the evaluation of the time loss to training and competition were taken from the medical report. To reduce recall bias, all team members were advised to record injuries immediately after occurrence. At the end of each season, recorded injuries were revised to make sure that the initial assessment of time loss was accurate.

FIS equipment regulations were introduced after the seasons 2000, 2003, 2007, 2012 and 2017 (seasons are displayed by the year in which the season ended). Because there was missing data from the 1999 season in the ISS of the Austrian Ski Federation the equipment regulation of 2000 could not be analysed. In 2017, only the geometry requirements of the giant slalom skis of the male racers was modified. The number of injuries using giant slalom skis was not sufficient to assess this equipment regulation. Four time periods remained to analyse the equipment regulations of 2003, 2007 and 2012. Period A included seasons 2001–2003 and period B seasons 2004–2007. The equipment regulation in 2012 was effective only for National-, A- and B-Team athletes. C-Team athletes were only entered in season 2014. Therefore, Period C lasted from season 2008–2012 for National-, A- and B-Team athletes and from season 2008–2013 for C-Team athletes. Consequently, period D lasted from season 2013–2017 for National-, A- and B-Team athletes and from season 2014–2017 for C-Team athletes. Due to the different durations of periods C and D, comparisons between periods B to C and periods C to D were separately performed for C-Team and non-C-Team athletes. Furthermore, the equipment regulation in 2012 did

not affect the slalom discipline. Therefore, a different period, C1, which did not include slalom injuries, was compared to period D.

Overall 17 seasons were analysed beginning with May 1, 2001 and ending with April 30, 2017. The number of team members of the Austrian Ski Federation fluctuated between 91 and 112 athletes per season, resulting in 1725 skier seasons (951 male and 774 female). 370 skier seasons were on the National Team, 357 on the A-Team, 536 on the B-Team and 462 on the C-Team. The qualification criteria for the different teams of the Austrian Ski Federation are publicly available.^{10,11} Qualification criteria are summarised and presented in supplementary Table 1.

It was distinguished between injury events and injuries because an injury event can result in several injuries to one or more body parts or body structures. Total severe injury events (TOT_{SIE}), events with severe knee injuries (KNEE_{SIE}) and events with severe ACL injuries (ACL_{SIE}) were analysed. Numbers for TOT_{SIE} contained all injury types, also knee and ACL injuries. KNEE_{SIE} included all kinds of severe knee injuries, including ACL injuries. An injury event was classified as severe if the time loss in training or competition was more than five weeks.

Injury incidence was expressed as number of injuries/100 athletes per season with their corresponding 95% Confidence Intervals (CIs). Risk ratios (RR) and their 95% CIs were calculated to compare injury incidence between consecutive periods (RR = incidence later period / incidence preceding period). Calculations were based on the Poisson model and used a Z-test (coefci, lmtest v0.9-37)¹² for comparing injury incidence. The difference between two compared injury incidences was significant, if the CI of a RR did not include the value 1. A two-tailed P-level of < 0.05 was considered statistically significant. Power analyses for non-significant differences were performed. Additionally, descriptive results were presented for subgroups differentiated by sex, competition, and training seasons. Due to low numbers, no inferential statistics were performed within these subgroups.

This study was conducted according to the 'ethical guidelines for surveys' approved by the Institutional Review Board of the Department of Sport Science as well as the Board for Ethical Issues of the University of Innsbruck. All athletes gave their written informed consent that anonymised data could be used for scientific purpose.

3. Results

Absolute numbers for skier seasons, TOT_{SIE}, KNEE_{SIE} and ACL_{SIE} in the different periods are presented for sex, competition, and training season in Table 1.

Fig. 1 depicts the seasonal TOT_{SIE} incidence for male and female athletes over 17 seasons. Mean incidence of TOT_{SIE} was 13.6/100 athletes per season (CI 11.4–16.1) for male and 20.2/100 athletes per season (CI 17.2–23.6) for female athletes.

For the comparison of consecutive periods, injury incidence and RR are presented in Table 2.

The comparison between periods A and B showed a significant increase ($p < 0.05$) of the TOT_{SIE} incidence after the equipment regulation in 2003. None of the other comparisons displayed statistical significance or statistical trends ($p < 0.1$). Power analyses for non-significant differences elicited type II errors greater than 20%. 60% of the calculated RR exceeded a value of 1. Four of the RR revealed an increase of injury incidence of 50% and more (RR between 1.50 and 1.63) after equipment regulations. Remarkably, in the group of National-, A- and B-Team 50% more ACL_{SIE} were recorded in the seasons after the equipment regulation in 2012. For C-Team athletes 63% more KNEE_{SIE} and 51% more ACL_{SIE} were observed after the equipment regulation in 2007. 40% of the calculated RR were below 1. The largest decrease of the injury incidence (RR 0.78) was

Table 1The number of skier seasons, TOT_{SIE}, KNEE_{SIE} and ACL_{SIE} for the different periods, separated by sex, competition, and training season.

	Period A				Period B				Period C/C1	Period C ^a			Period C1 ^b			Period D			
	Skier seasons	TOT _{SIE}	KNEE _{SIE}	ACL _{SIE}	Skier seasons	TOT _{SIE}	KNEE _{SIE}	ACL _{SIE}		Skier seasons	TOT _{SIE}	KNEE _{SIE}	ACL _{SIE}	TOT _{SIE}	KNEE _{SIE}	ACL _{SIE}	Skier seasons	TOT _{SIE}	KNEE _{SIE}
All																			
National-, A- and B-Team	225	24	20	14	305	51	35	21	367	77	39	25	59	32	20	366	48	37	30
C-Team	82	9	8	3	99	15	8	6	175	27	23	16	19	17	14	106	11	8	8
Total	307	33	28	17	404	66	43	27	542	104	62	41	78	49	34	472	59	45	38
Male																			
National-, A- and B-Team	125	12	9	4	178	24	14	8	207	40	18	14	33	17	13	202	25	20	16
C-Team	44	2	2	1	50	4	2	0	95	9	9	6	6	6	5	50	3	3	3
Total	169	14	11	5	228	28	16	8	302	49	27	20	39	23	18	252	28	23	19
Female																			
National-, A- and B-Team	100	12	11	10	127	27	21	13	160	37	21	11	26	15	7	164	23	17	14
C-Team	38	7	6	2	49	11	6	6	80	18	14	10	13	11	9	56	8	5	5
Total	138	19	17	12	176	38	27	19	240	55	35	21	39	26	16	220	31	22	19
Competition season ^c																			
National-, A- and B-Team	225	18	16	11	305	38	25	15	367	51	30	20	42	9	16	366	38	9	22
C-Team	82	5	4	1	99	11	6	4	175	23	20	14	18	3	13	106	6	3	4
Total	307	23	20	12	404	49	31	19	542	74	50	34	60	12	29	472	44	12	26
Training season ^d																			
National-, A- and B-Team	225	6	4	3	305	13	10	6	367	26	9	5	17	8	4	366	10	9	8
C-Team	82	4	4	2	99	4	2	2	175	4	3	2	1	1	1	106	5	4	4
Total	307	10	8	5	404	17	12	8	542	30	12	7	18	9	5	472	15	13	12

^a including Slalom for comparison to period B.^b excluding Slalom for comparison to period D.^c competition seasons lasted from the first competition in the season until 30th of April.^d training seasons lasted from 1st of May until the first competition.

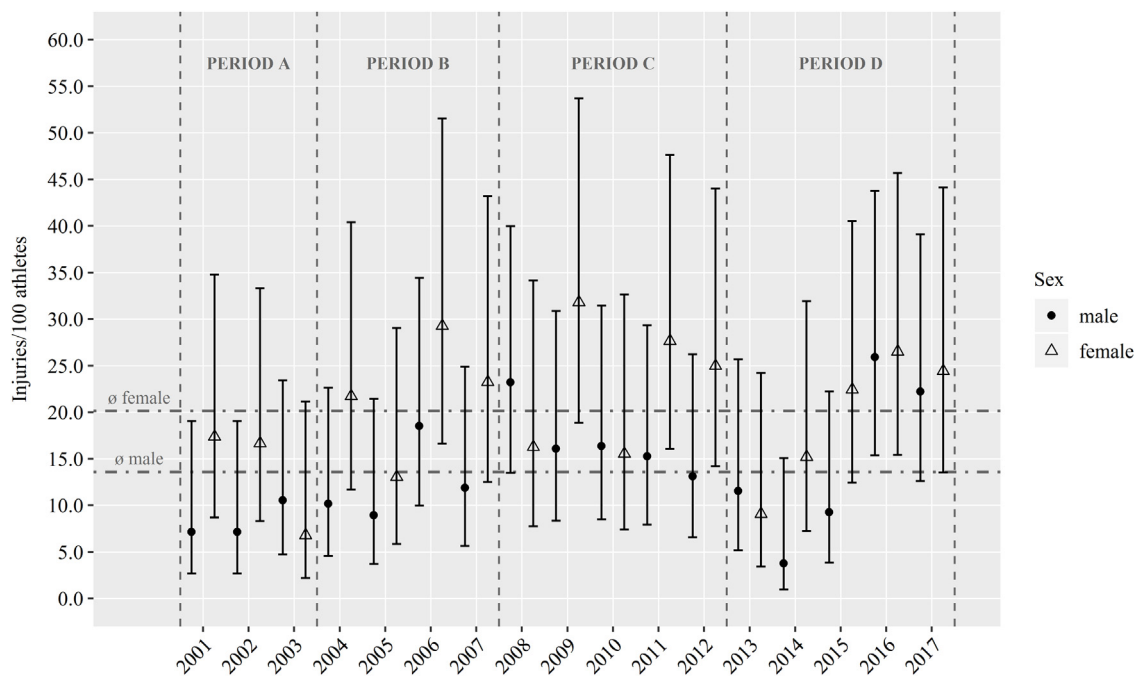


Fig. 1. TOT_{SIE} incidence and 95% CIs for male and female athletes of the Austrian Ski Federation (n = 1725). The dash dotted lines represent the means over 17 seasons. The dashed lines represent the introduction of new equipment regulations.

Table 2

Incidence and corresponding RR (95% CI in parentheses) for the comparison of TOT_{SIE}, KNEE_{SIE} and ACL_{SIE} in the different periods.

TOT _{SIE}				
Period	Incidence	Period	Incidence	RR
A (all Teams)	10.7 (7.6–15.1)	B (all Teams)	16.3 (12.8–20.8)	1.52 (1.00 ^a to 2.31) ^b
B (National-, A- and B Team)	16.7 (12.7–22.0)	C (National-, A- and B Team)	21.0 (16.8–26.2)	1.25 (0.88–1.79)
B (C-Team)	15.2 (9.1–25.1)	C (C-Team)	15.4 (10.6–22.5)	1.02 (0.54–1.91)
C1 (National-, A- and B Team)	16.1 (12.5–20.7)	D (National-, A- and B Team)	13.1 (9.9–17.4)	0.82 (0.56–1.19)
C1 (C-Team)	10.9 (6.9–17.0)	D (C-Team)	10.4 (5.7–18.7)	0.96 (0.45–2.01)
KNEE _{SIE}				
Period	Incidence	Period	Incidence	RR
A (all Teams)	8.9 (5.7–13.8)	B (all Teams)	11.5 (8.2–16.0)	1.17 (0.73–1.88)
B (National-, A- and B Team)	11.5 (8.2–16.0)	C (National-, A- and B Team)	10.6 (7.8–14.5)	0.93 (0.59–1.46)
B (C-Team)	8.1 (4.0–16.2)	C (C-Team)	13.1 (8.7–19.8)	1.63 (0.73–3.64)
C1 (National-, A- and B Team)	8.7 (6.2–12.3)	D (National-, A- and B Team)	10.1 (7.3–14.0)	1.16 (0.72–1.86)
C1 (C-Team)	9.7 (6.0–15.6)	D (C-Team)	7.5 (3.8–15.1)	0.78 (0.34–1.80)
ACL _{SIE}				
Period	Incidence	Period	Incidence	RR
A (all Teams)	5.5 (3.4–8.9)	B (all Teams)	6.7 (4.6–9.7)	1.21 (0.66–2.21)
B (National-, A- and B Team)	6.9 (4.5–10.6)	C (National-, A- and B Team)	6.8 (4.6–10.1)	0.99 (0.55–1.77)
B (C-Team)	6.1 (2.7–13.5)	C (C-Team)	9.1 (5.6–14.9)	1.51 (0.59–3.86)
C1 (National-, A- and B Team)	5.4 (3.5–8.4)	D (National-, A- and B Team)	8.2 (5.7–11.7)	1.50 (0.85–2.65)
C1 (C-Team)	8.0 (4.7–13.5)	D (C-Team)	7.5 (3.8–15.1)	0.94 (0.40–2.25)

^a exact lower limit of CI = 1.001.

^b significant difference (p < 0.05).

found for C-Team athletes with 22% less KNEE_{SIE} after the equipment regulation in 2012.

4. Discussion

The competition equipment was regulated to prevent injuries in alpine ski racing. Based on injury data of 17 seasons we assessed the equipment regulations of 2003, 2007 and 2012; to our knowledge, this is the first study to analyze these three FIS regulation changes.

Only one significant difference was found, namely a significantly higher incidence of TOT_{SIE} in the four seasons after the equipment regulation of 2003 compared to the three seasons before the regulation. In 2003 the side cut radius of downhill skis was increased

from 40 to 45 m for both sexes and the sole thickness of the ski boot was reduced from 50 mm to 45 mm for males. There was a substantial change regarding ski length. The minimum ski length of 155 cm (valid for all disciplines) was changed to 165 and 155 cm in slalom, 185 and 180 cm in giant slalom, 205 and 200 cm in super-G and 215 and 210 cm in downhill for males and females, respectively. Longer skis improve the terrain absorbing function of the front part of the ski,⁸ and increase the support area of the skier. Both aspects provide more stability for the skier and should lead to fewer falls and thus fewer injuries. However, it is unclear if the length regulation led to longer skis. Skiers who used skis that were longer than the minimum length before the equipment regulation may have switched to the new minimum length (which could be shorter). Either way, the results showed that the incidence for TOT_{SIE} was increased after the 2003 equipment regulation.

The equipment regulation in 2007 affected the minimum width of the ski waist, making skis wider in every discipline. The side cut radius for giant slalom skis was changed from 21 to 27 m and from 21 to 23 m for males and females, respectively. Standing height was further reduced by 5 mm for both sexes. Injury incidence showed no significant difference after this equipment regulation and no statistical trend was prevalent. Notably, in the period after this regulation the highest of all injury incidences was registered for the Austrian Ski Team. More than 20% of the National, A- and B-Team suffered a severe injury in each season.

Large changes to the ski requirements were made in 2012. Side cut radius and length were extended and maximum ski widths at waist and shovel were established for all disciplines except slalom. The most profound changes were made in giant slalom and super-G. For giant slalom skis, the side cut radius was increased from 27 to 35 m and from 23 to 30 m and for super-G skis, from 33 to 45 m and from 33 to 40 m for males and females, respectively. No significant difference and no statistical trends were found in the incidence of severe injuries after these changes. A recent study on Alpine World Cup ski racers also found no significant differences for severe injuries after the equipment regulation in 2012.³

With a power of less than 0.8 in the inference statistical analysis with non-significant results, the probability of detecting true effects was limited. However, the descriptive results showed that the calculated RR exceeded a value of 1 in the majority (60%) of the comparisons, indicating an increased injury incidence in the Austrian Ski Team after the equipment regulations (RR up to 1.63). When decreases of injury incidence for the Austrian Ski Team did occur after equipment regulations, the reductions were below 10% with the two exceptions of RR 0.78 and 0.82. Drawing conclusions may be limited by low statistical power, but neither the results of our inferential statistical analysis nor the descriptive empirical findings indicate, that the intended goal to reduce injuries was achieved through the equipment regulations.

There were limitations to this study. The study's sample was based on records from only one national ski federation, which questions the representativeness of the sample for FIS alpine ski racers and led to low sample sizes especially in the analysed subgroups. Consequently, all generalizations must be done with caution. However, in the context of the question of representativeness of the sample it is important to consider that the Austrian Ski Team is one of the largest and most professional teams participating in FIS races. The definition of a severe injury used by the Austrian Ski Federation may differ from other studies, but would not effect the comparisons made in our study between pre- and post equipment regulations. Furthermore, the slightly stricter definition of severe injuries would limit the comparability of the presented injury incidence only slightly; internal studies showed that only few cases of injuries in ski racing caused an absence of four weeks but not for five weeks. Due to the study's design, any findings were observational and not causal. Future equipment regulations have to be assessed comprehensively prior to their implementation. Computer simulations provide valuable insight. This has become evident for the terrain design of jumps^{13,14} and for the ACL injury risk during jump landing.^{15,16}

5. Conclusion

Even though statistical uncertainties remain, our findings allow the conclusion, that the implemented equipment regulations did

not cause a noticeable reduction of injuries. The three analysed equipment regulations were not appropriate or were counteracted by other factors.

Acknowledgements

The authors want to thank Carson Patterson MA for providing language help as well as athletes, coaches and medical staff of the Austrian Ski Federation for supporting this study.

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jsams.2020.07.005>.

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